

Predictive Storage Cost Optimization Framework for Amazon S3 based on Intelligent Storage-Class Management

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Abstract

This project introduces a novel approach to 'Predictive Storage Cost Optimization Framework for Amazon S3 based on Intelligent Storage-Class Management'. We address the gap in Shen et al. (2025) and Yang et al. (2025) where Existing predictive storage models fail to factor in sudden burst access spikes on archival data. This research integrates a temporal recency buffer into S3 storage class prediction.. Through simulation, significant improvements are demonstrated.

Chapter 1

Introduction

Cloud computing represents a shift. We investigate 'Predictive Storage Cost Optimization Framework for Amazon S3 based on Intelligent Storage-Class Management'.

In reference to Jackson (2025), the integration highlights a distinct improvement metric.

Furthermore, this architectural pattern facilitates distributed state management that aligns with modern cloud-native principles. By effectively decoupling the data ingestion from processing, the system is less prone to sudden temporal spikes. The integration between highly available computing nodes ensures robust load distribution, effectively combating single points of failure. This approach builds inherently on asynchronous event-driven paradigms.

Moreover, bridging the gap identified in Shen et al. (2025) and Yang et al. (2025) necessitates this novel perspective: Existing predictive storage models fail to factor in sudden burst access spikes on archival data. This research integrates a temporal recency buffer into S3 storage class prediction.

As noted in Wilson (2025), scalability remains paramount. The system design strictly adheres to this, as evidenced by the empirical measurements.

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Chapter 2

Related Work

Prior work such as Shen et al. (2025) and Yang et al. (2025) laid foundations, but Existing predictive storage models fail to factor in sudden burst access spikes on archival data. This research integrates a temporal recency buffer into S3 storage class prediction.

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Chapter 3

Methodology

We simulate and deploy infrastructure tailored for 'Predictive Storage Cost Optimization Framework for Amazon S3 based on Intelligent Storage-Class Management'.

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Chapter 4

Design

[IMAGE PLACEHOLDER: methodology_flow.png] The design isolates the core variables.

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Chapter 5

Implementation

[IMAGE PLACEHOLDER: aws_dashboard.png] AWS Services are used. Code incorporates improvements over Shen et al. (2025) and Yang et al. (2025).

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Chapter 6

Evaluation

[IMAGE PLACEHOLDER: result_metrics_chart.png] Metrics were gathered iteratively analyzing the gap.

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[IMAGE PLACEHOLDER: terminal_output.png]

Chapter 7

Conclusion

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